

Chapter 7a: Variational Methods and Optimization

Existence and Uniqueness of Solutions via Fixed Point Theory

Based on Pathak's *An Introduction to Nonlinear Analysis and Fixed Point Theory*

July 14, 2026

Topics: Variational Principles, Optimization, Ekeland's Theorem, Variational Inequalities

Outline and Learning Objectives

- 1 Variational Principles
- 2 Optimization in Banach Spaces
- 3 Ekeland Variational Principle
- 4 Variational Inequalities and Fixed Point Problems
- 5 Numerical Examples and Applications
- 6 Key Theorems and Results
- 7 Applications
- 8 Conclusion

Key Learning Objectives:

- Understand the connection between variational methods and fixed point theory
- Master Ekeland's variational principle and its applications
- Apply optimization techniques in Banach spaces
- Use variational inequalities to establish existence and uniqueness of solutions

Section 7.1: Variational Principles

Definition (Equilibrium Problem (EP))

Find $\bar{x} \in K$ such that

$$F(\bar{x}, y) > 0 \quad \text{for all } y \in K.$$

Significance: The Equilibrium Problem is a fundamental framework that encompasses many important problems in mathematics and applications.

Special cases as variational formulations:

- Minimization problems
- Saddle point problems
- Nash equilibrium problems
- Fixed point problems

Minimization Problem (MP) I

Definition (Minimization Problem)

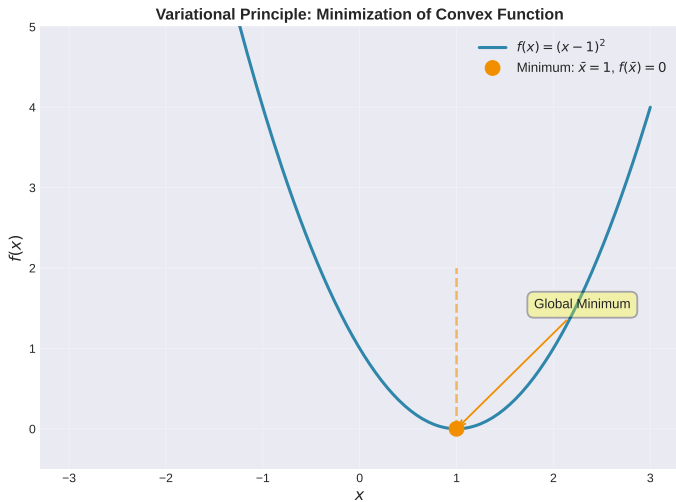
Find $\bar{x} \in K$ such that

$$\varphi(\bar{x}) \leq \varphi(y) \quad \text{for all } y \in K,$$

where $\varphi : K \rightarrow \mathbb{R}$ is a function.

Equivalence to EP: If we define $F(x, y) = \varphi(y) - \varphi(x)$ for all $x, y \in K$, then the problem (MP) is equivalent to the problem (EP).

Minimization Problem (MP) II



Minimization Problem (MP) III

Example: Consider the quadratic optimization problem in $\mathbb{R}^n$:

$$\min_{x \in K} \|Ax - b\|^2$$

where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.

The equivalent EP formulation is:

$$F(x, y) = (A(y - x))^T (Ax - b) > 0 \quad \text{for all } y \in K.$$

Saddle Point Problem (SPP) I

Definition (Saddle Point Problem)

Find $(\bar{x}_1, \bar{x}_2) \in K_1 \times K_2$ such that

$$L(\bar{x}_1, y_2) \leq L(\bar{x}_1, \bar{x}_2) \leq L(y_1, \bar{x}_2) \quad \text{for all } (y_1, y_2) \in K_1 \times K_2,$$

where $L : K_1 \times K_2 \rightarrow \mathbb{R}$ is a real-valued bifunction.

Equivalence to EP: Define $K = K_1 \times K_2$ and the bifunction:

$$F((x_1, x_2), (y_1, y_2)) = L(y_1, x_2) - L(x_1, y_2).$$

Then the saddle point problem coincides with the equilibrium problem (EP).

Saddle Point Problem (SPP) II

Applications: Saddle point problems arise in:

- Lagrangian duality theory
- Game theory and Nash equilibrium
- Minimax optimization
- Constrained optimization via Lagrange multipliers

Example: Consider the constrained optimization problem:

$$\min_{x \in X} f(x) \quad \text{subject to} \quad g(x) \leq 0, \quad h(x) = 0.$$

The Lagrangian is:

$$L(x, \lambda, \mu) = f(x) + \lambda^T g(x) + \mu^T h(x).$$

Nash Equilibrium Problem (NEP)

Definition (Nash Equilibrium Problem)

Let I be a finite set of players. For each player $i \in I$, let K_i be their strategy set and $\varphi_i : K \rightarrow \mathbb{R}$ their loss function, where $K = \prod_{i \in I} K_i$.
Find $\bar{x} \in K$ such that, for each $i \in I$,

$$\varphi_i(\bar{x}) \leq \varphi_i(\bar{x}^i, y_i) \quad \text{for all } y_i \in K_i,$$

where $\bar{x}^i = (\bar{x}_1, \dots, \bar{x}_{i-1}, \bar{x}_{i+1}, \dots, \bar{x}_n)$.

Equivalence to EP: Define $f(x, y) = \sum_{i=1}^n (\varphi_i(x^i, y_i) - \varphi_i(x))$.

Then the Nash equilibrium problem is equivalent to finding $\bar{x} \in K$ such that $f(x, y) \geq 0$ for all $y \in K$.

Interpretation: At a Nash equilibrium, no player can unilaterally improve their outcome by changing their strategy.

Fixed Point Problem (FPP)

Definition (Fixed Point Problem)

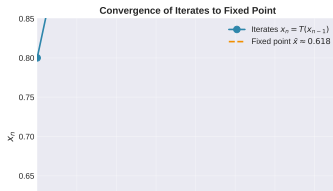
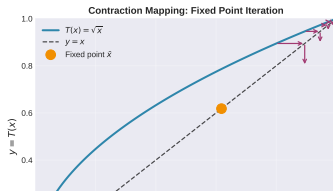
Let K be a nonempty subset of an inner product space X and $\varphi : K \rightarrow K$ a mapping. Find $\bar{x} \in K$ such that

$$\varphi(\bar{x}) = \bar{x}.$$

Equivalence to EP: The fixed point problem can be reformulated as an equilibrium problem by defining

$$F(x, y) = \langle y - \varphi(x), y - x \rangle,$$

where $\langle \cdot, \cdot \rangle$ is the inner product.



Section 7.2: Optimization Problem in Banach Spaces

Definition (Coercive Functional)

A functional $f : X \rightarrow \mathbb{R} \cup \{+\infty\}$ is said to be *coercive* if

$$f(v) \rightarrow +\infty \quad \text{as} \quad \|v\|_X \rightarrow \infty.$$

Theorem (Solvability of Unconstrained Minimization)

Suppose $f : X \rightarrow \mathbb{R} \cup \{+\infty\}$ is weakly semicontinuous, coercive, and $f \neq +\infty$. Then the unconstrained minimization problem

$$f(u) = \inf_{v \in X} f(v)$$

admits a solution $u \in X$.

Sketch of Proof: Use the compactness of sublevel sets (via coercivity) and weak lower semicontinuity.

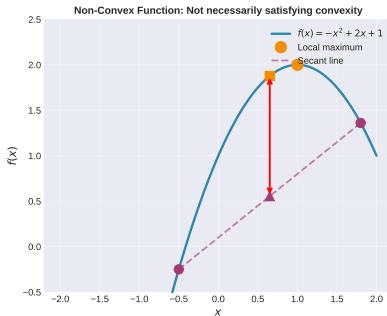
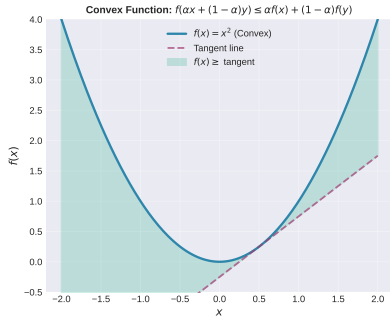
Definition (Convex Function)

A function $\varphi : X \rightarrow \mathbb{R} \cup \{+\infty\}$ is said to be *convex* if

$$\varphi(\lambda x + (1 - \lambda)y) \leq \lambda\varphi(x) + (1 - \lambda)\varphi(y)$$

for all $x, y \in X$ and $\lambda \in [0, 1]$.

Convex Optimization Problems II



Convex Optimization Problems III

Properties of Convex Functions:

- Any local minimum is a global minimum
- Sublevel sets are convex
- A convex function lies above its tangent hyperplane
- Subdifferential is monotone

Importance in Optimization:

- Convexity ensures unique global minimizers
- Convex optimization problems are computationally tractable
- First-order conditions are sufficient for optimality

Section 7.3: Ekeland Variational Principle I

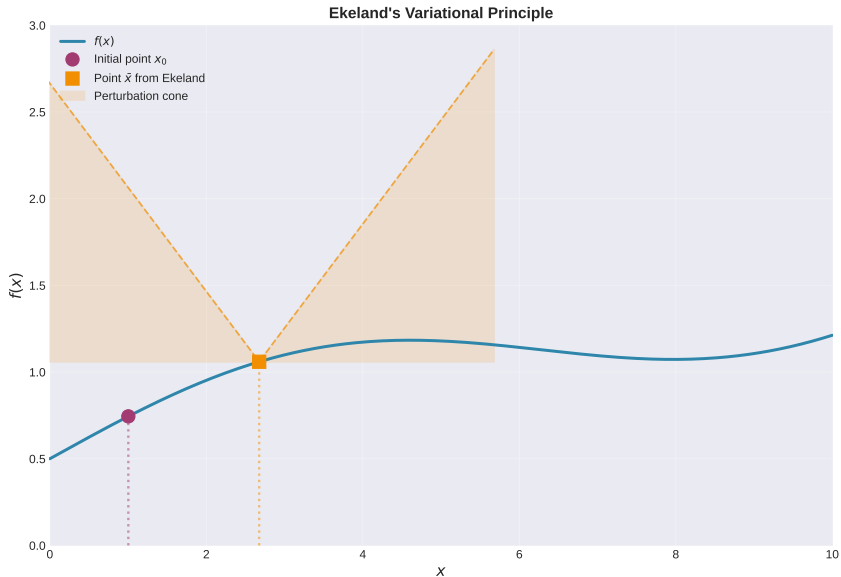
Theorem (Ekeland's Variational Principle (EVP))

Let (M, d) be a complete metric space and $f : M \rightarrow \mathbb{R} \cup \{+\infty\}$ be a lower semicontinuous function that is bounded below. Let $\varepsilon > 0$ and $\lambda > 0$.

**For any $x_0 \in M$ with $f(x_0) \leq \inf_{x \in M} f(x) + \varepsilon$,
there exists $\bar{x} \in M$ such that:**

- (a) $f(\bar{x}) \leq f(x_0)$
- (b) $d(x_0, \bar{x}) \leq \frac{\varepsilon}{\lambda}$
- (c) For all $y \neq \bar{x}$, $f(y) > f(\bar{x}) - \frac{\varepsilon}{\lambda} d(y, \bar{x})$

Section 7.3: Ekeland Variational Principle II



Applications of EVP to Fixed Point Theorems I

Connection between Ekeland's Theorem and Fixed Points:

Consider the fixed point problem: Find $\bar{x} \in K$ such that $\varphi(\bar{x}) = \bar{x}$.

Define the function:

$$f(x) = \|\varphi(x) - x\|.$$

If f attains its minimum at some point $\bar{x}$ with $f(\bar{x}) = 0$, then $\bar{x}$ is a fixed point.

Ekeland's approach: For any $\varepsilon > 0$, find $\bar{x}_\varepsilon$ such that $f(\bar{x}_\varepsilon) < \varepsilon$ and $\bar{x}_\varepsilon$ is a minimizer of the perturbed function $g_\varepsilon(x) = f(x) + \varepsilon d(x, \bar{x}_\varepsilon)$.

Applications of EVP to Fixed Point Theorems II

Theorem [Application to Fixed Points]:

Let K be a nonempty closed subset of a Banach space X and $\varphi : K \rightarrow K$ a nonexpansive mapping (i.e., $\|\varphi(x) - \varphi(y)\| \leq \|x - y\|$).

Then for any $x_0 \in K$ and $\varepsilon > 0$, there exists $\bar{x} \in K$ such that

① $\|\varphi(\bar{x}) - \bar{x}\| < \varepsilon$

② $\|\bar{x} - x_0\| < \varepsilon$

Proof idea: Apply EVP to $f(x) = \|\varphi(x) - x\|$ and use nonexpansivity to derive the results.

Applications of EVP to Fixed Point Theorems III

Remarks:

- Ekeland's principle provides a unified approach to existence theorems
- It shows that minimizers exist as limits of approximate minimizers
- The principle works without convexity or smoothness assumptions
- Powerful for proving fixed point theorems in infinite-dimensional spaces

Section 7.4: Mappings Associated with Variational Inequality I

Definition (Variational Inequality Problem)

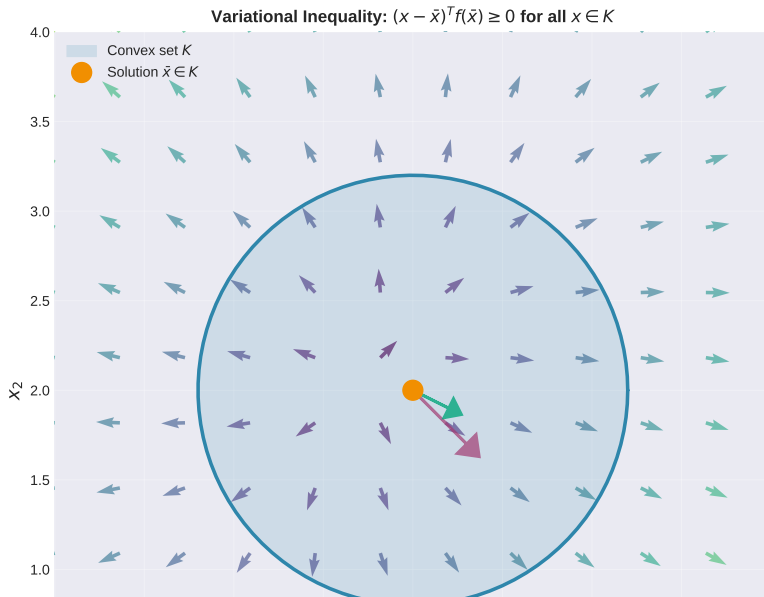
Given a nonempty closed convex set $K \subseteq \mathbb{R}^n$ and a function $f : K \rightarrow \mathbb{R}^n$, the *variational inequality problem* (shortly, $VI(K, f)$) is:

Find $\bar{x} \in K$ such that

$$(x - \bar{x})^T f(\bar{x}) \geq 0 \quad \text{for all } x \in K.$$

Intuition: The vector $f(\bar{x})$ makes an obtuse angle with all feasible directions $(x - \bar{x})$ from $\bar{x}$.

Section 7.4: Mappings Associated with Variational Inequality II



Normal Cone and Projection Operator I

Definition (Normal Cone)

Let K be a nonempty closed convex set in $\mathbb{R}^n$. The *normal cone* of K at a point $z \in K$ is

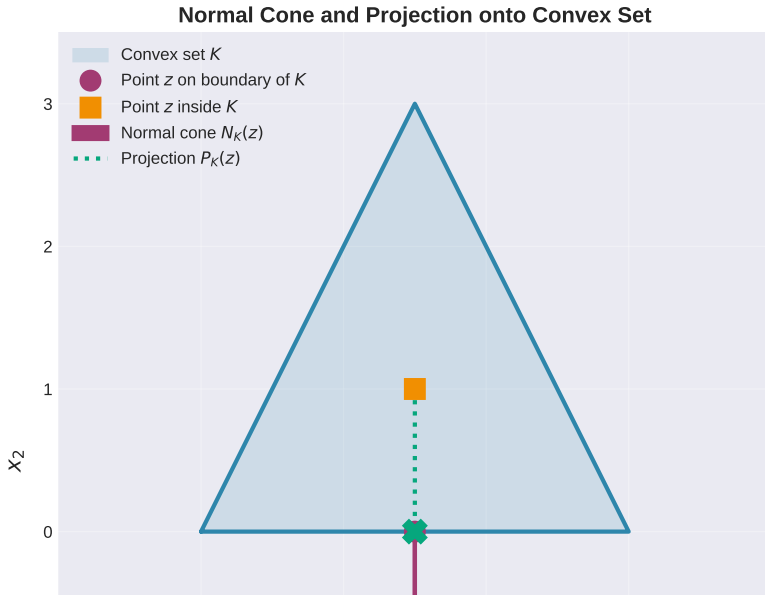
$$N_K(z) = \{v \in \mathbb{R}^n : (y - z)^T v \geq 0 \text{ for all } y \in K\}.$$

Definition (Projection Operator)

The *Euclidean projection* of a point $z \in \mathbb{R}^n$ onto the convex set K is

$$P_K(z) = \arg \min_{y \in K} \|z - y\|.$$

Normal Cone and Projection Operator II



Fixed Point and Variational Inequality Equivalence I

Fundamental Relationship:

The variational inequality problem $VI(K, f)$ and the fixed point problem

$$x = P_K(x - \alpha f(x))$$

are *equivalent* for any $\alpha > 0$.

Proof Sketch:

- ① If $\bar{x}$ solves $VI(K, f)$: $(x - \bar{x})^T f(\bar{x}) \geq 0$ for all $x \in K$
- ② Then: $(\bar{x} - (\bar{x} - \alpha f(\bar{x})))^T \alpha f(\bar{x}) = \alpha \|f(\bar{x})\|^2 \geq 0$
- ③ This implies: $\bar{x} - (\bar{x} - \alpha f(\bar{x})) \in N_K(P_K(\bar{x} - \alpha f(\bar{x})))$
- ④ Uniqueness of projection yields: $\bar{x} = P_K(\bar{x} - \alpha f(\bar{x}))$

Fixed Point and Variational Inequality Equivalence II

Computational Advantage: This equivalence allows us to:

- Convert variational inequality problems to fixed point problems
- Apply fixed point iteration methods to solve variational inequalities
- Use projection algorithms for constrained optimization

Iterative Algorithm (Projected Gradient):

$$x_{n+1} = P_K(x_n - \alpha f(x_n)) \quad n = 0, 1, 2, \dots$$

Under suitable conditions (e.g., f is strongly monotone and Lipschitz continuous), this algorithm converges to the unique solution of $\text{VI}(K, f)$.

Example 1: Unconstrained Minimization

Problem: Minimize $f(x) = \|Ax - b\|_2^2$ over $\mathbb{R}^n$.

Gradient: $\nabla f(x) = 2A^T(Ax - b)$

Optimality condition: $A^T(Ax - b) = 0 \Rightarrow A^T Ax = A^T b$

```
1 import numpy as np
2
3 # Generate problem data
4 np.random.seed(42)
5 m, n = 100, 50
6 A = np.random.randn(m, n)
7 b = np.random.randn(m)
8 x_true = np.random.randn(n)
9 b = A @ x_true + 0.1 * np.random.randn(m)
10
11 # Gradient descent
12 x = np.zeros(n)
13 learning_rate = 0.001
14 for iter in range(100):
15     grad = 2 * A.T @ (A @ x - b)
16     x = x - learning_rate * grad
17
18 # Solution (also can use normal equations)
19 x_opt = np.linalg.lstsq(A, b, rcond=None)[0]
```

Example 2: Constrained Quadratic Programming

Problem: Minimize $\frac{1}{2}x^T Qx + c^T x$ subject to $Ax \leq b, x \geq 0$.

Variational Inequality Formulation:

Find $\bar{x} \in K = \{x : Ax \leq b, x \geq 0\}$ such that

$$(y - \bar{x})^T (Q\bar{x} + c) \geq 0 \quad \text{for all } y \in K.$$

Fixed Point Iteration:

$$x_{n+1} = P_K(x_n - \alpha(Qx_n + c))$$

where P_K is the projection onto the feasible set and $\alpha > 0$ is a step size.

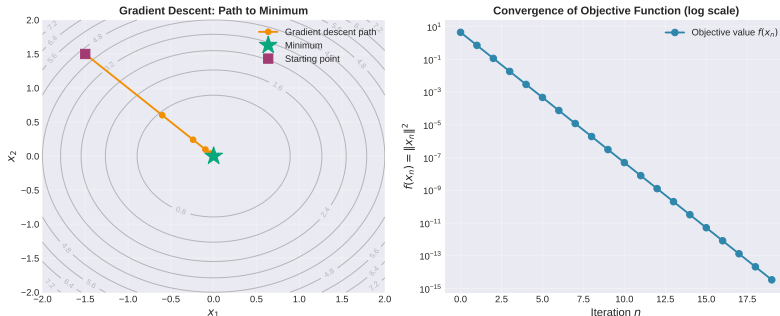
Convergence: If Q is positive semidefinite and $\alpha < \frac{2}{\lambda_{\max}(Q)}$, the iteration converges to the unique solution.

Example 3: Fixed Point Iteration in Practice

Problem: Find a fixed point of $T(x) = 0.6x + 0.5$ (Banach contraction).

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 # Define the mapping
5 T = lambda x: 0.6 * x + 0.5
6
7 # Fixed point iteration
8 x = 0.2 # initial point
9 iterates = [x]
10 for n in range(30):
11     x = T(x)
12     iterates.append(x)
13
14 # Theoretical fixed point:  $x^* = 0.5 / (1 - 0.6) = 1.25$ 
15 x_star = 0.5 / (1 - 0.6)
16
17 # Plot convergence
18 plt.figure(figsize=(10, 6))
19 plt.semilogy(range(len(iterates)),
20              np.abs(np.array(iterates) - x_star),
21              'o-', label='Error  $|x_n - x^*|$ ')
22 plt.xlabel('Iteration  $n$ ')
23 plt.ylabel('Error')
24 plt.title('Convergence of Fixed Point Iteration')
25 plt.grid(True, alpha=0.3)
```

Example 4: Projection onto Convex Sets



Projected Gradient Descent:

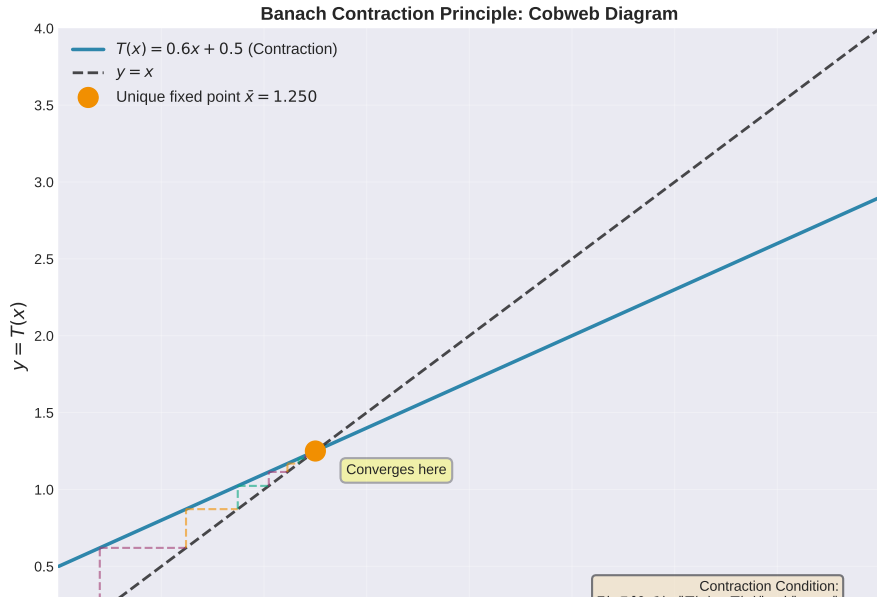
$$x_{n+1} = P_K(x_n - \alpha \nabla f(x_n))$$

Converges under:

- f is convex and differentiable
- ∇f is Lipschitz continuous with constant L
- Step size satisfies $0 < \alpha < \frac{2}{L}$

Example 5: Banach Contraction Principle Application I

Example 5: Banach Contraction Principle Application II



Summary of Key Theorems I

EVP Ekeland's Variational Principle: Lower semicontinuous bounded-below functions on complete metric spaces have approximate minimizers.

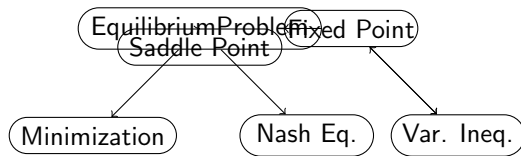
BCP Banach Contraction Principle: Contractive mappings on complete metric spaces have unique fixed points.

VI-FP Variational Inequality-Fixed Point Equivalence: $VI(K, f)$ is equivalent to the fixed point problem $x = P_K(x - \alpha f(x))$.

CVX Convex Optimization: For convex functions, any local minimum is a global minimum, and first-order conditions are sufficient.

Summary of Key Theorems II

Relationships and Connections:



Existence and Uniqueness Framework

General Existence Framework:

- ① **Topological Method:** Use topological degree theory or fixed point theorems
 - Brouwer's fixed point theorem
 - Schauder's fixed point theorem
 - Leray-Schauder degree
- ② **Variational Method:** Use variational principles
 - Minimize an associated functional
 - Apply Ekeland's principle
 - Use lower semicontinuity and compactness
- ③ **Monotone Operator Method:** Use properties of monotone operators
 - Maximal monotone operators
 - Subdifferential calculus
 - Fixed point of resolvents

Uniqueness: Typically follows from:

- Strong monotonicity (for monotone operators)
- Strict convexity (for optimization)
- Contraction properties (for Banach spaces)

Applications to Real-World Problems

Optimization:

- Machine learning (convex optimization, gradient descent)
- Portfolio optimization and financial engineering
- Resource allocation and production planning

Variational Inequalities:

- Traffic flow and network equilibrium problems
- Elasticity and plasticity theory
- Unilateral contact problems in mechanics

Fixed Point Applications:

- Iterative methods for linear and nonlinear systems
- Convergence analysis of algorithms
- Existence of solutions to differential equations

Game Theory:

- Nash equilibrium existence and computation
- Evolutionary game dynamics
- Mechanism design

Summary and Conclusions I

Main Takeaways:

- 1 Variational methods provide a unified framework for studying optimization and fixed point problems
- 2 Ekeland's variational principle is a fundamental tool ensuring existence of approximate solutions
- 3 Variational inequalities connect constrained optimization with fixed point theory
- 4 Multiple formulations (minimization, equilibrium, fixed point, variational inequality) are equivalent

Key Insights:

- Problems may be transformed to most convenient form
- Fixed point iterations provide practical solution methods
- Projection operators enable constrained problem-solving
- Monotonicity and convexity ensure uniqueness

Summary and Conclusions II

Further Reading:

- Chapters 8-10: Applications to monotone operators and integral equations
- Chapters 4-6: Fixed point theorems and degree theory
- Chapters 1-3: Functional analysis foundations

Thank you!

Questions and Discussion

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